

SPECIAL CONDITIONS OF
CONTRACT (SCC), SCOPE OF
WORK (SOW) & TECHNICAL
SPECIFICATIONS

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SPECIAL CONDITIONS OF CONTRACT (SCC)

1. GENERAL:

- 1.1 Special conditions of contract shall be read in conjunction with the general conditions of contract, specification of work, drawing and any other document forming part of this contract the context so requires. Where any portion of the general conditions of contract is repugnant to or at variance with any provisions of the special conditions of contract, then unless a different intention appears, the provision(s) of the general conditions of contract only to the extent that such repugnance of variations in the special conditions of contract.
- 1.2 Whenever it is mentioned in the specification that the contractor shall perform certain work or provide certain facilities, it is understood that the contractor shall do at their own cost and the contract price shall be deemed to have included cost of such performance and provisions, so mentioned. The materials, design and workmanship shall satisfy the applicable relevant Indian standards, the job specification contained herein & codes referred to. Where the job specifications stipulate requirements in additions to those contained in the standard codes and specifications, these additional requirements shall also be satisfied.
- 1.3 The intending bidders should be deemed to have visited the site and familiarized themselves thoroughly with the site conditions before submitting their bid. Non-Familiarity with the site condition will not be considered a reason either for extra claims or for not carrying out work in strict conformity with drawing and specification.
- 1.4 Bidders are advised to visit the sites (Sectionalizing Valve Station- 19, 19A, 20, 24, 25 and 32) and acquaint themselves fully about the locations details & quantum of job needed to be executed at respective site prior to bid submission without any financial implication to GAIL regarding visit which they may require for preparation and submitting the bid. Claims and objection due to ignorance of the existing conditions or inadequacy of information will not be considered after submission of the bid and during implementation.
- 1.5 The work is to be carried out in “Restricted area” and successful bidders will have to work as per instructions of GAIL and shall follow GAIL’s safety guidelines. GAIL shall not be responsible for any accident/damage caused to the men and material of the contractor.

- 1.6 The jobs covered under the “**Scope of Work**” of the Contract are to be completed in all respect at the earliest, without compromising on quality.
- 1.7 After the completion of job at respective region, vendor has to submit the completion report, in the form of MOM signed by EIC.
- 1.8 In the absence of any specification covering any, material, design of work(s) the same shall be performed / supplied / executed in accordance with Standard Engineering Practice as per the instructions / directions of the Engineer-In-Charge (EIC), which will be binding on the Contractor.
- 1.9 All the work shall be carried out as per relevant Indian Standards, specifications of Batteries, in compliance with the standard rules and as directed by the EIC.
- 1.10 The offered equipment shall be brand new with state-of-the-art technology and a proven field track report in LPG, gas processing and petrochemical industry only. No prototype equipment shall be offered.

2. SITE DETAILS:

The details of the site locations are given in the scope of work are as follows:

SV	ADDRESS
SV-19	MOTA JAMPURA, BANASKANTHA
SV-19A	LAKHAW, PATAN
SV-20	LAXMIPURA, BANASKANTHA
SV-24	RAJPURIYA, BANASKANTHA
SV-25	KALIMATI, BANASKANTHA
SV-32	BARWA, PALI

Site-In-Charge shall be nominated to look after the installations and commissioning of the Battery Banks.

3. PAYMENT TERMS:

3.1 For Supply Portion (SUPPLY OF MATERIALS)

- a. 80% of the supply payment shall be released after material delivery & acceptance.
- b. Balance 20% of the supply payment will be released after successful completion installation & commissioning in all aspects and acceptance by EIC.
- c. Vendor is required to update invoice in GAIL's Sarthi / VIM portal and payment shall be released after 15 working days of the submission of invoice.

3.2 Service Part

The charges quoted for installations & commissioning will be on lump sum and will include all expenses related to the installation & commissioning job including To & Fro charges, lodging, boarding etc. GAIL will not pay any amount other than what is quoted in respective SOR item. 100% payments shall be made after successful dismantling, Installation & Commissioning and collection of the scrap from site and handing over the system to GAIL duly certified by SIC/EIC. Interested party may visit the site before submission of their offer for work estimation or any other clarification. In case any clarifications, same may be asked before bidding.

Note: There is only one-line item in GeM, so bidder may quote price considering supply part, service part and buyback of scrap batteries. Successful bidder will submit detailed price break up for making internal PO in our SAP system.

4. CONTRACTOR'S RESPONSIBILITIES:

General tools and tackles required will be arranged by the Contractor. The contractor shall provide, operate and maintain all the necessary tools, tackles and other equipment.

4.1 All lifting tools & tackles, man, material/machines, Drilling/Hydra/Crain/JCB etc. required for execution of the jobs safely mentioned in the scope of work is in the scope of contractor.

4.2 The normal working hours is eight hours a day except Sunday and holidays with a break of half an hour for lunch. Working hours shall be notified from time to time and may be changed as per exigencies of work.

- 4.3 All material, manpower, consumables, transportation, lodging & boarding, loading & unloading etc. required to complete the jobs mentioned in the scope of work shall be in the scope of the contractor.
- 4.4 The contractor shall have to depute at least one experienced supervisor who does have experience in the field of electrical system & installation of such battery banks. He/she should be able to manage site activities while executing the job at site and guide their personnel to carry out the work as and when required as per the instructions of the Engineer-In-charge. No extra payment will be made for providing supervisor in order to achieve desired workmanship and quality; the contractor shall also deploy experienced persons for the execution of work.
- 4.5 The Contractor shall at their own expense arrange for safety provisions as required by the Engineer-in-charge, in respect of all facilities in connection therewith.
- 4.6 **SITE SAFETY PROCEDURES:** The site safety procedures are the guidelines to Contractors and other person who are involved in the activities at GAIL, regarding the safe working requirement to be observed when working at site.
- 4.7 **DUTIES AND RESPONSIBILITIES:** These site safety procedures will be followed without prejudice to any statutory requirement and guidelines that may be in force at regulations shall be observed during the execution of work.
- 4.8 **WORK PERMIT:** Any job inside the plant premises must be carried out under the work permit. Different types of works permits are Hot Work permit, Cold Work permit, vessel entry, excavation, vehicle entry, electrical lock out and work at height permit etc.
- 4.9 **FIRE PREVENTION REGULATIONS:** The fire and safety regulations should be strictly observed within the plant premises by all persons for prevention of fire/accident.
- 4.10 **PERSONNEL PROTECTIVE EQUIPMENT (PPEs):** The contractors shall provide all personnel with proper protective clothing and equipment as per job

requirement. The PPEs shall be checked by Fire & Safety Department of GAIL before use.

- 4.11 ACCIDENT REPORTING: Contractors should report all accident/incident (if any) to EIC without fail whether minor or major.
- 4.12 HOUSE KEEPING: Contractor must ensure to keep their respective areas clean at all times and well in order.
- 4.13 The Contractor shall acquaint himself with access availability, facilities, local labor etc.
- 4.14 The procurement and supply in sequence and at the appropriate time of all be entirely be contractor's responsibility. Contractor shall not use any of the equipment or materials issued to him by Owner for temporary works, manufacturing erection aids etc. Misuse of materials will be seriously viewed and deduction at penal rates will be made from the Contractors bill for such quantities that are misused.
- 4.15 Contractor should produce necessary ESIC code before commencement of work or workmen who are not covered under ESI act, the contractor should take appropriate workmen compensation insurance policy and submit a copy of the same to GAIL (India) Limited.
- 4.16 Contractor should produce necessary EPF code before commencement of work.
- 4.17 Contract Price is deemed to be inclusive of all expenses towards above responsibilities.
- 5. CONTRACT COMPLETION PERIOD: Total contract period shall be 05 MONTHS to be reckoned from the date of LOA/PO. This includes design, supply, installation, testing, commissioning and handling over the system to GAIL' respective site.
- 6. GATEPASS/IDENTITY CARD: All the items of Contractor required to complete the job will be through Gate Pass & entries made on Security Gate. The contractor shall give the details & confirmation of this in written. All the Man-Power of Contractor required the valid Gate Pass along with Identity Card as per the formalities of Security Department.

7. **DEFECT LIABILITY PERIOD:** All supply & work shall be warranted against material defect and poor workmanship, for a period of at least ONE YEAR after successful completion of the entire job in all respect as declared by respective EIC/SIC.
8. **DOCUMENTATION/REPORT:** The contractor will maintain all records and relevant documents and shall submit the same to respective EIC/SIC for verification. Contractor shall submit two copies (soft & hard) of site report of complete work, as-built drawings and other documents as per scope of work in proper format. List of documents to be submitted to GAIL while supplying the material at site is as given below:
 - a) Three copies of as-built drawings, specifications, details, test certificates and other relevant certificates for supplied materials with one soft copy.
 - b) Three Copies of O&M manual with one soft copy.
 - c) Manufacturer Warranty certificate.
 - d) Factory Test Certificate/Report.
9. **NO DAMAGE TO INSTALLATION:** Contractor shall ensure that during the progress of the work no damage shall occur to installations due to any reason, if so happens, contractor shall have to repair/replace the same at their own risk and cost. The decision of EIC shall be final and binding on the contractor in this case.
10. **CARE OF WORKS:** From the commencement to completion of the work, the contractor shall take full responsibility for the care for entire installation including all temporary works and in case of damage, losses or injury which may happen to the work or to a part thereof or to any worker from any cause the cost of such loss or damage will be recovered from the contractor's bill.
11. **PUNITIVE FINES:**
 - 11.1 Punitive fines on contractors are imposed for repetitive violation of safety rules & regulations during execution of jobs. Objective of punitive fines is to work as deterrent for contractors in violation of safety rules & regulation and to improve safety atmosphere in general at all site.
 - 11.2 For first time violation of safety rules & regulation by any contractor, HOD (F&S) will issue a warning letter to contractor with intimation to OIC of work center and EIC.
 - 11.3 In case of second time violation of safety rules & regulations by same contractor, OIC will call contractor in person and will have a meeting to discuss reason for repetitive violation along with EIC and HOD (F&S). A warning letter will also be issued by EIC to contractor.

In case of further violation, punitive fines will be imposed on contractor. Amount of fine will be decided as per severity of violation of safety. However, minimum fine would be Rs.10,000/- and in multiple of Rs.10,000/-, thereafter. This will be limited to 5% of contract value, as maximum cumulative penalty.

1.0 GENERAL INFORMATION

SCOPE OF SUPPLY:

(a) One set of Battery Bank Consists of

- (i) Supply of 24V NI-CD Battery Bank (consisting of 19 cells each cell rating 1.2V, 1195 AH.) Connection arrangement 19 cells to be connected in series. For each Battery Bank there will be 19 cells of 1.2V, 1195 AH.
- (ii) Supply of one set of Mandatory spares (consists of 2 nos. of cells complete with leads & accessories) as mentioned in point no. (i)
- (b) Installation, testing & Commissioning of battery banks consists of dismantling, Erection & Commissioning of one no. of Existing battery bank (38 cells).

The bidder should quote considering all the items mentioned above for Installation, testing & Commissioning of battery banks.

- (c) Party has to buy back the old battery. Buy back is mentioned in clause no- 07 in subsequent page. The Buyback price should be clearly indicated in their offer (terms & condition) separately. (Total no. of 228 old cells.)

Rating of Existing Batteries/ buy back the old batteries:

- (a) No. of Batteries: 228 (Each cell rating 1.2V, 1195 Ah)
- (b) No. of Cells per bank: 19, total no. of Ni-Cd Battery Bank
- (c) Battery Type: KPL1195P
- (d) Make: SNPS
- (e) Model: KPL
- (f) Battery MFG year: 2000

Interested party may visit the site before submission of their offer for work estimation or any other clarification. In case any clarifications, same may be asked before bidding.

2.0 DETAILED SCOPE:

- (a) UNIT OF MEASUREMENT 1 EA = One complete bank consisting of 19 Ni - Cd BATTERIES with all accessories containing frame, cell interconnectors etc. and any other items required for complete banks including 2 no's of spare batteries along with their accessories.
- (b) Total 228 nos. of cells are being procured for 12 sets of battery banks including 24 nos. of spare cells as 12 sets of mandatory spares. The battery banks are to be complete along with stand, cell interconnectors, electrolyte, hardware and other accessories as mentioned in detailed specification.
- (c) The complete shifting of batteries to the battery room from the storage place & shifting of old battery banks to the identified place is in the scope of the party.
- (e) The agency has to completely remove the existing old 12 sets of battery banks with its accessories in the respective battery units & erect the newly supplied battery banks & associated accessories in complete in place of the old battery bank.
- (f) Complete erection of the battery bank is to be carried out by the agency along with the accessories.
- (g) Batteries shall preferably be located in the single tier arrangement.
- (h) As battery is the most critical system, the erection & commissioning work shall be taken up on topmost priority & shall be completed within the allotted time frame.
- (i) The commissioning of the newly installed battery bank is in the scope of the party. Party has to carry out all charge, discharge tests for the newly commissioned battery bank and handover the complete system to GAIL.
- (j) The old batteries with accessories are to be taken back by the party on buy back basis.

Party has to submit Declaration for disposal of old batteries as per statutory

rules as batteries being e-waste.

- (k) Standard items like instruction cards, apron, goggles, cards, gloves etc. are to be supplied by the agency at free for cost along with each battery bank.
- (l) Mandatory spares are to be supplied along with each battery bank as indicated below:

-> 2 no. of cells (1195 AH, 1.2 V) along with all accessories like ventilation caps, cover plates, interconnecting links, cable lugs, hardware, electrolyte etc. per battery bank.
- (m) All works & materials involved for complete supply and successful erection & commissioning of the battery banks are in the scope of agency.
- (n) Party has to submit the complete layout drawing, test reports & instruction manuals along with O&M manuals to site after commissioning of each battery bank in book format as well as in soft copy.
- (o) Commissioning of each battery at site shall only be carried out either by the battery manufacturer himself or under the supervision of the battery manufacturer.
- (p) Interested party may visit the site before submission of their offer for work estimation or any other clarification. In case any clarifications, same may be asked before bidding.

3.0 BATTERY RATINGS

- (a) Battery Bank Voltage 24V
- (b) No. of Cells per bank 19 (each cell rating 1.2V, 1195 Ah)
- (c) Battery type Stationary Nickel -Cadmium Pocket Plate Low discharge type (KPL)
- (d) Capacity for five (5) hour rate for each bank 1195 AH
- (e) Nominal discharge voltage per Cell 1.2
- (f) Float voltage 1.42V/Cell

4.0 CODES AND STANDARDS

4.01 All standards, specifications and codes of practice referred to herein, shall be the latest editions including all applicable official amendments and revisions as on date of opening of techno-commercial bid.

In case of conflict between this specification and those (IS codes, Standards etc.) referred to herein, the former shall prevail. All works shall be carried out as per the following standards and codes:

IEC 60623 / IS 10918 – Specification for vented type Nickel Cadmium Batteries.

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IS 1069 Quality tolerances for water for storage batteries

IEC 60993 – Electrolyte for vented Nickel-Cadmium cells

Indian Electricity Acts

4.02 Equipment complying with other internationally accepted standards such as IEC., BS, VDE etc. will also be considered if they ensure performance and constructional features equivalent or superior to standards listed above. In such a case, the Bidder shall clearly indicate the standard(s) adopted, furnish a copy in English of the latest revision of the standards along with copies of all official amendments and revisions in force as on date of opening of techno-commercial bid and shall clearly bring out the salient features for comparison.

5.0 GENERAL TECHNICAL REQUIREMENT

5.01 Equipment

- (a) DC Batteries shall be stationary Nickel Cadmium Pocket plate type (KPL)/ conforming to IS 10918. The batteries shall be low discharge performance type as specified. For the purpose of design an ambient temperature of 50 degree centigrade and relative humidity of 85% shall be considered.
- (b) DC batteries shall be suitable for standby duty. The batteries shall normally be permanently connected to the load in parallel with a charger and shall supply the load during emergency conditions when AC supplies are lost. Batteries shall be suitable for a long life under continuous float operations and occasional discharges. The batteries shall be boost charged at about 1.54 to 1.7 volts per cell maximum and float charged at about 1.42 V/cell.
- (c) The number of cells in one battery bank for the 24 Volts shall be 19 (& two

no's of battery bank in parallel, each cell rating 1.2V, 1195Ah).

- (d) Batteries should be suitable for continuous operation for the maximum ambient temperature as defined in technical parameters.

5.02 Construction Features

(a) Containers

Containers shall be made of polypropylene plastic material. Containers shall be robust, heat resistance, leak proof, nonabsorbent, alkali resistant, non-bulging type and free from flaws, such as wrinkles, cracks, blisters, pin holes etc. Electrolyte level lines shall be marked on container in case of translucent containers.

(b) Vent Plugs

Vent plugs shall be provided in each cell. They shall be anti-splash type, having more than one exit hole shall allow the gases to escape freely but shall prevent alkali from coming out. The design shall be such that the water loss due to evaporation is kept to minimum. In addition, the ventilator shall be easily removed for topping up the cells and of such dimensions that the syringe type hydrometer can be inserted into the vent to take electrolyte samples.

(c) Plates

The plates shall be designed for maximum durability during all service conditions including high rate of discharge and rapid fluctuations of load. The construction of plates shall conform to latest revisions of IS 10918. The separators shall maintain the electrical insulation between the plates and shall allow the electrolyte to flow freely. Separators should be suitable for continuous immersion in the electrolyte without distortion.

The positive and negative terminal posts shall be clearly marked.

(d) Sediment Space

Sufficient sediment space shall be provided so that cells will not have to be cleaned during normal life and prevent shorts within the cells.

(e) Electrolyte

The electrolyte shall be prepared from battery grade potassium hydroxide conforming to IEC 60993. The cells can be shipped either in charged condition or in dry condition. Necessary electrolyte for make-up shall be supplied separately. Necessary electrolytes for spare cells are also required to be supplied separately.

(f) Connectors and Fasteners

Nickel plated copper connectors shall be used for connecting adjacent cells and PVC insulated flexible copper cables shall be used for inter-row / inter-tier / inter-bank connections. Bolts, nuts and washers shall be Stainless Steel / Nickel coated steel to prevent corrosion. The thickness of Nickel coating of connectors should be not less than 0.02 mm. All the terminals and cells inter-connectors shall be fully insulated or have insulation shrouds. End take off connections from positive and negative poles of batteries shall be made by single core cables having stranded AL conductors and XLPE insulation (by employer). Necessary supports and lugs for termination of these cables on batteries shall also be supplied by the contractor. All connectors and lugs shall be capable of continuously carrying the 30 minutes discharge current of the respective batteries and through fault short circuit current which the battery can produce and withstand for the period declared. Contractor shall furnish necessary sizing calculations to prove compliance to the same. Suitable number of Inter-rack connectors shall be supplied by the Bidder to suit the battery room layout during detailed engineering.

(g) Battery racks

Mild steel racks for all the batteries shall be provided. They shall be free standing type mounted on porcelain/hard rubber/PVC pads insulators/High impact plastic insulators. Batteries shall preferably be located in the single tier arrangement. The batteries racks and supports for cable termination shall be coated with three (3) coats of antialkali paint of approved shade. Nameplates, resistant to alkali, for each cell shall be attached on to the necessary racks. The bottom of the tier of the stand shall not be less than 150 mm above the floor. Wherever racks are transported in dismantled conditions, match markings shall be provided to facilitate easy assembly.

(h) Manufacturer's Identification System

The following information shall be indelibly marked on outside of each cell.

-> Manufacturers' name and trade marks

-> Country and year of manufacture.

-> Manufacturer type designation.

-> AH capacity at 5-hour discharge rate.

-> Serial number

(i) Additional Information

The following information shall be given on the instruction cards supplied with the battery:

-> Manufacturer's instructions for filling and initial charging of the battery together with starting and finishing charging rate.

-> Maintenance instructions.

-> Designation of cell in accordance with IS 10918.

-> Storing conditions of electrolyte.

6.0 TESTS

All equipment to be supplied shall be of type tested design. The contractor shall submit for Owner's approval the reports of all the type tests as listed in this specification and carried out within last ten years from the date of techno-commercial bid opening. These reports should be for the test conducted on the equipment similar to those proposed to be supplied under this contract and the test(s) should have been either conducted at an independent laboratory or should have been witnessed by a client. However, if the contractor is not able to submit report of the type test(s) conducted within last ten years from the date of techno-commercial bid opening, or in the case of type test report(s) are not found to be meeting the specification requirements, the contractor shall conduct all such tests under this contract at no additional cost to the owner either at third party lab or in presence of GAIL representative and submit the reports for approval. All acceptance and routine tests as per the specification and relevant standards shall be carried out. Charges for these shall be deemed to be included in the equipment price. The type test reports once approved shall be treated as reference. An endorsement sheet will be furnished by the manufacturer confirming similarity and "No design change". Minor changes if any shall be highlighted on the endorsement sheet.

The Contractor shall submit for Owner's approval, the reports of all the type tests carried out as per latest IS-1146 (for all applicable tests for containers) / IS-10918 (for Ni-Cd batteries).

6.02 Routine and Acceptance tests shall be as per QA & Inspection table of battery.

(a) Commissioning Checks:

All tests as listed below shall be carried out by the agency at site after completion of installation.

-> Physical Examination

-> MARKING

-> Polarity and absence of short circuit.

-> Ampere - hour capacity

-> Insulation resistance

-> The Contractor shall arrange for all necessary equipment, including the variable resistor, tools, tackles and instruments.

(b) Party has to carry out the interface cable connection work & newly supplied battery bank using existing cables. Any other items which are required for interfacing work shall be supplied at free cost by the party.

7.0 Buy Back Policy

Party has to take back the complete existing 12 sets of battery banks along with accessories on buy back basis. Party has to submit Declaration for disposal of old batteries in accordance to the Battery Management & Handling rules 2001 or latest as batteries being e-waste.

(a) Each set of existing NiCd Battery Bank Consists of:

(i) 19 cells in a single battery bank hence total 228 number of cells in 12 banks, each cell rating 1.2V, 1195Ah.

(ii) Existing accessories that will be replaced with new ones in 12 set of battery banks

(b) Dismantling of existing Battery Bank is in the scope of vendor. The bidder should quote considering all these activities.

(c) Party has to buy back the old battery. The Buyback price should be taken into

consideration while submitting their offer. (Total no of 228 old cells.)

Further, the tender shall stipulate the following procedure to be followed by GAIL and vendor/contractor:

1. That old materials are to be handed over to the vendor/contractor in terms of the contract with proper documentation on “as is where is basis”. The vendor/contractor is required to take away such materials out of GAIL immediately after handing over to them.
2. That the vendor/contractor will not sell these materials to any GAIL employee without written permission of HR Department- GAIL, to be obtained by the GAIL employee(s). Even after permission is obtained by employee(s) from GAIL, the Contractor is free to take decision whether to sell such item (s) to GAIL employee(s) or not.
3. That vendor/contractor will accept payment only by Cheque/Demand Draft for sale of old item(s) to GAIL employee(s).
4. That vendor/Contractor will maintain a separate record for such sales to GAIL employees and will make the same available as and when requested by GAIL.